

# Precision, accuracy and uncertainty

Working scientifically  
Both papers · Combined and Separate  
10 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                                                                           | ANSWER                                                           | ✓                        |
|----|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------|
| 1  | Define precision.                                                                                                  | How close repeated measurements are to each other.               | <input type="checkbox"/> |
| 2  | Define accuracy.                                                                                                   | How close a measurement is to the true value.                    | <input type="checkbox"/> |
| 3  | Define the resolution of a measuring instrument.                                                                   | The smallest change the instrument can detect.                   | <input type="checkbox"/> |
| 4  | Define uncertainty.                                                                                                | The interval within which the true value can be expected to lie. | <input type="checkbox"/> |
| 5  | State how the mean of a set of repeated readings is calculated.                                                    | Add up the readings, then divide by the number of readings.      | <input type="checkbox"/> |
| 6  | State what is meant by the range of a set of readings.                                                             | The smallest and largest readings, e.g. from 1.20 s to 1.25 s    | <input type="checkbox"/> |
| 7  | State how uncertainty is calculated from a set of repeated readings.                                               | Uncertainty = (largest reading – smallest reading) ÷ 2           | <input type="checkbox"/> |
| 8  | Describe how a final result should be written when its uncertainty is known.                                       | As mean ± uncertainty, with a unit, e.g. 1.22 ± 0.03 s.          | <input type="checkbox"/> |
| 9  | Three readings are close together but far from the true value. Describe them using the words precise and accurate. | They are precise but not accurate.                               | <input type="checkbox"/> |
| 10 | State what repeated measurements help you to judge: precision or accuracy.                                         | Precision                                                        | <input type="checkbox"/> |